Symlit_Arrows_Study3

OSF: https://osf.io/q825d

Internet_Archive: https://archive.org/details/osf-registrations-q825d-v1

1) Data collection. Have any data been collected for this study already?

Yes, pilot data. Please question 8.

2) Hypothesis. What's the main question being asked or hypothesis being tested in this study?

We investigate young children's comprehension of deictic reference in the graphic domain. Three-, and four-year-old children are presented with an object-choice-task in which a cartoon agent is providing cues to help participants choose between two options presented at the left and right bottom of the screen. The cues are asymmetric drawings that may be read as pointing either left or right by virtue of having an opening, bump, dent, overlay on either side (cf. illustration).

For the purpose of our main analyses, we code as correct when children choose the option that the symbols are directed to. We are using age as continuous predictor in a between-subjects design and investigate when children's group performance deviates from chance level. We hypothesize that children's performance will surpass chance level after 48 months of age.

3) Dependent variable. Describe the key dependent variable(s) specifying how they will be measured.

The key dependent variable is children's choice behavior in the object-choice-task. Choice items presented on the left and right side of the screen. Children respond by pointing to the side of their choice. Children's choices are coded as correct (1) or incorrect (0) in line with the logic of the task**.**

4) Conditions. How many and which conditions will participants be assigned to?

Children will be presented with an equal number of trials of all types of stimuli. Types of stimuli are not treated as different conditions (see analyses below).

5) Analyses. Specify exactly which analyses you will conduct to examine the main question/hypothesis.

All analyses will be computed in R (R Core Team, 2018). Bayesian models will be run in Stan (http://mc-stan.org/) and implemented via the function brm of the package brms (Bürkner, 2017). We will use logistic Bayesian generalized linear mixed models (GLMM) to fit children's responses (0/1) as a function of their absolute age in days. We use default priors and include trial and sex as fixed effects to be controlled for. Age will be added as a random slope within item type. Trial number will be added as a random slope within subject.

Model: **condition*z.age +z.trial +z.sex +(age|itemtype)+(z.trial|id)**

- correct: correct choice (0/1);
- z.age: age in days, centered to a mean of 0
- z.trial: trial number, scaled
- z.sex: participants' sex (male/female), scaled
- itemtype: factor specifying the type of cue (cf. illustration)
- id: participant id

The analysis will model participants binary choices to predict the probability of children interpreting the cues correctly and model how this probability will change as a function of their absolute age in days. In order to evaluate the relevance of age for children's performance, we will compare a full model as specified above with a reduced model lacking age by using WAIC scores and weights (McElreath, 2016). Furthermore, we will inspect the model estimates for the different predictors (including their 95% Credible Interval (CrI)). The criterion for settling when children perform above chance is the point at which the 95% CrI for a particular trajectory does no longer overlap with a midline demarcating the 50% chance level.

To further explore our data, we also bin participants by year of age. To test whether group-level performance is above chance in all age groups, we use two-tailed one-sample t-tests with the chance level set to .5. We provide Cohen's d as a standardized effect size for significance testing (computed via the function `cohensD`). Additionally, each participant's data will also be submitted to a binomial test to determine whether their performance is above chance on an individual level.

6) Outliers and Exclusions. Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Prior to the main study phase, children are tested with familiarization trials in which they simply have to point to the position of an item that was hidden in plain sight. This serves to ensure that children are comfortable and understand the task.

In order to enter the main study phase, children should have a success rate of 66% in the familiarization. The experimenter provides at least four trials. If three in four are correct (75%), the child proceeds to the main phase. If not, two additional familiarization trials are run. If the child is correct in four out of six trials (66%), she proceeds to the main study phase. If not, two additional trials are presented. If six in eight are correct (75%), the child enters the main study phase. If children do not fulfill the criterion in a maximum of eight trials, they are dropped from the sample.

Children will also be excluded from the sample if they do not provide at least 10 valid out of 20 possible trials in the main study phase. Trials are excluded if children do not respond, if responses cannot be interpreted, if parents or caregivers interfere with the procedure, or on the basis of technical problems or experimenter error.

7) Sample Size. How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

In case children do not perform above chance as a group within this age-range, the age-range will be extended by a year increasing the sample by 24 children until performance surpasses chance-level. If children are succeeding already at 36 months, we will extend the age-range to 24 months.

8) Other. Anything else you would like to pre-register?

We ran a pilot version with small samples (N = 4) of four- and five-year-old children (Total N = 8) to evaluate the age-appropriateness of the task and our stimulus design. We document all pilots and the specific conclusions we drew from them in the supplemental material accompanying the manuscript. Data collection for the main sample started three days prior to preregistration. None of the data of the main sample have been sighted at the time of preregistration.

9) Name. Give a title for this AsPredicted pre-registration

Symlit_Arrows_Study3:unconventionalarrows

10) **Illustration of Stimuli**

